

Problem-Solution Fit

Date	10 Jun 2026
Team ID	xxxxxxx
Project Name	Smart Crop
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Primary: Smallholder and commercial farmers seeking to maximize crop yield metrics and input performance. Secondary: Agricultural researchers, regional policymakers, and agronomical stakeholders studying interaction footprints.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Financial Boundaries: Fragmented budgets lacking allocation space for expensive laboratory chemical soil profiling tests. Device Access: Primary interaction occurs via basic smartphones or intermittent network connectivity zones out in rural fields.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] TRADITIONAL INSTINCT & MANUAL ESTIMATION Pros: Immediate, zero upfront financial cost, rooted in localized generational heritage. Cons: Highly inaccurate under climate shifting, fails to track macro chemical degradation (N, P, K). STATIC LAB SOIL AUDITING Pros: Highly precise chemical accuracy.
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		Cons: Extremely slow, expensive, and fails to combine dynamic macro climate parameters like rainfall trends
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Sub-optimal Yield Rates: Low productivity outcomes arising from incorrect crop selection matches (Encountered every planting cycle). Resource Waste: Guesswork-driven fertilizer application rates causing nutrient degradation or financial loss (Continuous occurrence). Policy Misalignment: Regional strategy maps lacking evidence-based environmental parameters (Encountered every seasonal cycle).	9. PROBLEM ROOT / CAUSE [RC] Complex Multi-Variable Equations: Crop optimization demands synthesizing 7 parameters (\$N, P, K, pH\$, temperature, humidity, rainfall), which cannot be calculated manually. Asymmetric Data Silos: Live weather parameters are separated from chemical soil telemetry, leaving gaps in context.	7. BEHAVIOR + ITS INTENSITY [BE] Reactive Countermeasures: Farmers heavily over-apply fertilizers when visible growth markers decline (High operational intensity). Static Planning: Planning relies on historical calendar assumptions rather than responsive real-time variables (Moderate intensity).
3. TRIGGERS TO ACT [TR] Experiencing sharp drop-offs in field harvest volumes compared to previous tracking cycles. Spikes in market fertilizer procurement costs forcing smart conservation workflows. Policymakers needing empirical proof to justify financial	10. YOUR SOLUTION [SL] OptiCrop Production Optimization Engine Farmers: Instant multi-variable yield matching based on soil (\$N, P, K, pH\$) and local climate inputs.	8. CHANNELS OF BEHAVIOR [CH] ONLINE ONLINE CHANNELS Mobile responsive web application panels, direct weather network feeds, and cloud macro reporting suites. OFFLINE CHANNELS

distribution programs.	Assessments: Dynamic suitability grading gauges mapping alternative crop compatibility paths. Researchers: Open historical analytical reporting suites to drive macro-level planning.	Local agricultural extension outreach clinics, village community kiosks, and physically logged soil sampling cards.
4. EMOTIONS BEFORE / AFTER [EM] Emotions Before Using OptiCrop: Anxiety and operational stress caused by volatile seasonal rainfall parameters and crop selection uncertainty. Frustration over rising fertilizer costs and unpredictable field output trends. Emotions After Using OptiCrop: Confidence and peace of mind driven by empirical, data-driven optimization calculations. Empowerment through actionable control over soil resource health and regional strategy matrices.		OFFLINE